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Original Research Article

Biodegradation of low-density polyethylene (LDPE) and polyethylene terephthalate (PET) by indigenous bacterial isolates from dump yard soils

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Abstract

Background: Plastic pollution is a critical environmental issue due to the persistence and recalcitrance of polymers such as low-density polyethylene (LDPE) and polyethylene terephthalate (PET). Microbial biodegradation offers an eco-friendly solution for plastic waste management.

Aim: To isolate, identify, and evaluate the biodegradation potential of indigenous plastic-degrading bacteria from dump yard soils.

Materials and Methods: Soil samples were collected from two plastic-contaminated dumping sites. Bacterial isolates were obtained using standard microbiological techniques and screened on Minimal Salt Medium supplemented with LDPE. Identification was performed through morphological, biochemical, and molecular characterization, including 16S rRNA gene sequencing. Biodegradation efficiency was assessed using weight-loss analysis of PET films. Comparative evaluation was carried out between individual isolates and synergistic bacterial consortia.

Results: Four isolates exhibited significant plastic-degrading activity. PET showed higher degradation (37.5%) compared to LDPE (27.5%). Bacterial consortia demonstrated enhanced efficiency, achieving 61.25% degradation for PET and 46.25% for LDPE, outperforming individual isolates.

Conclusion: Indigenous bacterial consortia possess significant potential for biodegradation of synthetic plastics and offer a sustainable approach for plastic waste management.

Keywords: 16S rRNA gene sequencing, Dump yard soil bacteria, Low-density polyethylene (LDPE), Microbial consortia, Plastic biodegradation, Polyethylene terephthalate (PET).

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1. Introduction

Plastic pollution has emerged as a critical global environmental issue owing to the exponential increase in plastic production, extensive consumer usage, and inadequate waste management practices. Since their large-scale commercialization in the mid-twentieth century, synthetic plastics have become indispensable in packaging, construction, healthcare, textiles, and consumer goods. Global plastic production has exceeded 390 million tonnes annually, a substantial proportion of which accumulates in landfills, dump yards, and natural ecosystems due to low recycling rates and improper disposal practices.[1] The persistence of plastic waste in terrestrial and aquatic environments poses serious ecological, economic, and public health concerns.

Among the various synthetic polymers, low-density polyethylene (LDPE) and polyethylene terephthalate (PET) are extensively used because of their durability, lightweight nature, chemical resistance, and cost effectiveness. LDPE is commonly employed in carry bags, packaging films, and agricultural mulch, while PET is widely used in beverage bottles, food containers, and textile fibers. However, these same physicochemical properties render both polymers highly resistant to natural degradation processes, leading to their prolonged persistence in the environment for decades to centuries.[2] Conventional plastic disposal and management strategies such as landfilling, incineration, and mechanical recycling suffer from several limitations, including incomplete degradation, generation of toxic by-products, high energy consumption, and the formation of secondary pollutants such as microplastics.[3]

Dump yards and landfill sites represent major sinks for plastic waste and constitute unique ecological niches characterized by long-term exposure to synthetic polymers, fluctuating environmental conditions, and limited nutrient availability. These conditions exert selective pressure on resident microbial communities, promoting the adaptation of microorganisms capable of colonizing plastic surfaces and potentially utilizing polymers or their additives as carbon and energy sources. Surface weathering caused by ultraviolet radiation, temperature fluctuations, and mechanical stress may further enhance microbial attachment by introducing functional groups onto polymer surfaces, thereby facilitating subsequent biological degradation.

Microbial biodegradation of plastics has gained increasing attention as an environmentally sustainable alternative to conventional waste management approaches. Several bacteria and fungi have been reported to degrade

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synthetic polymers through processes involving surface colonization, biofilm formation, oxidative modification, and enzymatic depolymerization.[4] Bacterial genera such as *Bacillus*, *Pseudomonas*, *Streptomyces*, *Rhodococcus*, *Nocardia*, and *Micrococcus* have been frequently associated with polyethylene degradation, whereas PET degradation has been intensively studied following the discovery of *Ideonella sakaiensis*, which produces PETase and MHETase enzymes capable of hydrolyzing ester bonds in the PET polymer matrix.[5]

Although numerous studies have reported plastic-degrading microorganisms, many investigations rely predominantly on laboratory-adapted strains, qualitative screening methods, or single-strain systems that may not accurately reflect natural biodegradation processes. In contrast, microbial consortia often demonstrate superior degradation efficiency due to synergistic interactions, complementary enzymatic activities, and sequential metabolism of degradation intermediates.[6] Indigenous microorganisms isolated from plastic-contaminated dump yard soils are therefore of particular interest, as they are naturally adapted to polymer-rich environments and may exhibit enhanced degradation capabilities under nutrient-limited conditions.

In this context, the present study aims to isolate and characterize indigenous plastic-degrading bacteria from dump yard soils and to evaluate their biodegradation potential against LDPE and PET under controlled laboratory conditions. By combining morphological, biochemical, and 16S rRNA gene-based molecular identification with quantitative weight-loss assays, this study seeks to provide a comparative assessment of LDPE and PET biodegradation by individual bacterial isolates and synergistic consortia. The findings contribute to a deeper understanding of microbial plastic degradation and highlight the potential application of native bacterial communities in sustainable plastic waste management strategies.[7]

2. Materials and Methods

2.1. Sample collection

Soil samples were collected from two different dumping yards located in Hyderabad with the history of dumping plastic wastes from a decade ago. Samples were taken from the upper 5–10 cm of the soil layer where visible accumulation of plastic debris was observed. Approximately 100 g of soil was collected from each site using sterile spatulas and stored in sterile polypropylene containers. The containers were properly labeled with date and location of collection. All samples were transported to the laboratory in aseptic conditions and stored at 4°C until further analysis.

2.2. Isolation of plastic-degrading bacteria

After collecting soil samples from four different locations within the dump yard, the samples were processed for bacterial isolation using the standard microbial techniques. One gram of soil from each site was suspended in sterile saline to prepare the initial dilution, followed by tenfold serial dilutions up to the required range. From each dilution, 0.1 mL was aseptically spread onto nutrient agar plates to obtain well-separated bacterial colonies. All plates were incubated in an inverted position at 30–37°C for 24–48 hours. Individual colonies were carefully selected from each plate and streaked onto fresh nutrient agar to obtain pure cultures. These pure isolates were further maintained on nutrient agar slants for subsequent screening of Plastic degrading activity, morphological, biochemical and molecular characterization.

2.3. Screening for plastic-degrading activity

Primary screening of all bacterial isolates for plastic-degrading potential was performed in Minimal Salt Medium (MSM) broth supplemented with 0.5% (w/v) LDPE powder, where each isolate was inoculated into the MSM–LDPE broth and incubated at 30°C for 5–7 days under shaking conditions. Isolates showing visible signs of LDPE utilization—such as turbidity changes, reduction in LDPE particle density, or surface clearing—were considered preliminary degraders. For quantitative evaluation, a weight-loss assay was conducted using pre-weighed, sterile PET (polyethylene terephthalate) film strips introduced into MSM broth inoculated with the screened bacterial cultures. Parallel uninoculated control flasks were maintained to account for abiotic weight changes. After 30 days of incubation at 30°C under shaking conditions, PET films were retrieved, washed thoroughly with 2% SDS to remove adhered biomass, rinsed with sterile water, and dried at 50°C to constant weight. The percentage weight reduction was calculated using the formula:

where Wo and Wt represent initial and final weights of the plastics, respectively.

2.4. Identification of plastic-degrading bacteria

Pure bacterial isolates obtained from dumpyard soil were subjected to microbiological, biochemical and molecular identification. Initially, isolates were cultured on nutrient agar and incubated at 30–37°C for 24 h to ensure purity. Colony morphology, including size, colour, margin, elevation, and surface characteristics, was recorded, followed by microscopic examination after Gram staining to determine cell shape and Gram reaction. Biochemical characterization was carried out using standard tests, including catalase, oxidase, indole production, methyl red, Voges–Proskauer, citrate utilization, urease, nitrate reduction, starch hydrolysis, and gelatin liquefaction. All biochemical assays were performed using freshly grown cultures to obtain preliminary genus-level identification.

2.5. Molecular identification

Genomic DNA of each selected isolate was extracted using a commercial bacterial DNA isolation kit (Qiagen). The purity and concentration of DNA were checked using a Nanodrop spectrophotometer at 260/280 nm. The 16S rRNA gene was amplified by PCR using universal bacterial primers 27F (5′-AGAGTTTGATCMTGGCTCAG-3′) and 1492R (5′-TACGGYTACCTTGTTACGACTT-3′). PCR amplification was carried out in a 25 µL reaction mixture containing template DNA (50 ng), primers (0.5 µM each), dNTPs (0.2 mM), Taq polymerase (1 U), and 1× PCR buffer. The cycling conditions included an initial denaturation at 95°C for 3 min; followed by 35 cycles of 95°C for 30s, 55°C for 30s, and 72°C for 90s; with a final extension

at 72°C for 10 min. The PCR products were resolved on 1% agarose gel and visualized under UV illumination. Purified PCR products were subjected to Sanger sequencing, and the resulting sequences were edited and assembled to generate consensus sequences. Species-level identification was achieved by comparing the 16S rRNA gene sequences with reference sequences available in the NCBI GenBank database using the BLASTn algorithm. Isolates showing sequence similarity were assigned to the closest related species, while phylogenetic relationships were inferred where necessary to confirm taxonomic placement using molecular evolutionary genetic analysis (MEGA) (ref).

2.6. Bacterial extract preparation and fractionation

To obtain cell-free extracts, each bacterial isolate was cultured in nutrient broth at 30°C under shaking conditions (150 rpm) for 72 h. Cells were harvested by centrifugation at 10,000 rpm for 15 min at 4°C, and the resultant supernatant was retained as the extracellular fraction. The cell pellets were resuspended in 50 mM phosphate buffer (pH 7.0) and subjected to ultrasonication (10 cycles of 30 s pulses with 30 s intervals on ice) to obtain cell-free intracellular extracts. Debris was removed by centrifugation at 12,000 rpm for 20 min at 4 °C, and the clarified supernatant was filter-sterilized through a 0.22 µm membrane. Total protein concentration in each extract was determined by the Bradford assay using bovine serum albumin as the standard.

For fractionation, the cell-free extracellular extract was subjected to sequential solvent partitioning. Equal volumes of the extract were partitioned successively with n-hexane, ethyl acetate, and n-butanol (1:1 v/v) using a separating funnel. Each organic fraction was collected, dried over anhydrous sodium sulfate, and evaporated to dryness under reduced pressure using a rotary evaporator at 40°C. The resulting residues were reconstituted in dimethyl sulfoxide (DMSO) at a stock concentration of 10 mg/mL and stored at −20°C until further use. The aqueous residual fraction was lyophilized and similarly reconstituted. Each fraction was subsequently evaluated for enzymatic and polymer-degrading activities to determine the most active fraction responsible for observed biodegradation.

2.7. Fourier-transform infrared (FTIR) spectroscopic analysis

To confirm chemical modifications in plastic polymer surfaces following microbial degradation, FTIR spectroscopic analysis was performed on both treated and untreated LDPE and PET film samples. After the 30-day biodegradation assay, retrieved polymer films were thoroughly cleaned with 2% SDS, rinsed with distilled water, and air-dried. FTIR spectra were recorded using an attenuated total reflectance (ATR) accessory on a Perkin-Elmer Spectrum 100 FTIR spectrometer over a wavenumber range of 400–4000 cm^{-1} with a resolution of 4 cm^{-1}. A total of 32 scans were co-added per spectrum. Spectral data were analysed using Perkin-Elmer Spectrum software, and changes in absorption bands corresponding to carbonyl (C=O, ~1715 cm^{-1}), hydroxyl (O–H, ~3200–3500 cm^{-1}), and ester (C–O–C, ~1250 cm^{-1}) functional groups were monitored to assess oxidative and hydrolytic modifications indicative of biological degradation.

2.8. Scanning electron microscopy (SEM)

Surface morphological changes in LDPE and PET films following microbial degradation were examined by scanning electron microscopy. Post-incubation polymer film samples were fixed in 2.5% glutaraldehyde in 0.1 M phosphate buffer (pH 7.2) for 2 h at 4°C, dehydrated through a graded ethanol series (50%, 70%, 90%, and 100%), and critical-point dried. Dried specimens were mounted on aluminium stubs and sputter-coated with gold-palladium alloy to a thickness of approximately 10 nm under vacuum. Samples were examined using a JEOL JSM-6490LV scanning electron microscope at an accelerating voltage of 15 kV. SEM images were acquired at magnifications ranging from 500× to 5000× to document the extent of surface erosion, pit formation, biofilm attachment, and structural degradation relative to uninoculated control samples.

2.9. Enzyme activity assays

To investigate the enzymatic basis of plastic degradation, the cell-free extracellular fractions of all four isolates were assayed for laccase, lipase, and protease activities using standardised spectrophotometric methods. Laccase activity was measured using 2,2′-azino-bis(3-ethylbenzothiazoline-6-sulfonic acid) (ABTS) as substrate; the oxidation of ABTS was monitored at 420 nm (ε = 36,000 M^{-1} cm^{-1}) and expressed as U/mL, where one unit is defined as the amount of enzyme oxidising 1 µmol of ABTS per minute. Lipase activity was determined by titrimetric assay using olive oil emulsion as substrate; released free fatty acids were titrated against 0.05 N NaOH with phenolphthalein as indicator, and activity was expressed as µmol fatty acid released per minute per mL. Protease activity was assayed using 1% azocasein as substrate; the reaction mixture was incubated at 37°C for 30 min, terminated with 10% trichloroacetic acid, and the absorbance of the soluble fraction was measured at 440 nm. All enzyme assays were performed in triplicate and enzyme activities were normalised to total protein content (specific activity, U/mg protein). Uninoculated buffer served as the blank control in all assays.

3. Results

3.1. Morphological, microscopic, and biochemical characterization of bacterial isolates

Multiple morphologically distinct bacterial colonies were isolated from dump yard soil samples, indicating a diverse microbial population. Only a few isolates showed plastic-degrading ability, among which four Gram-positive bacterial isolates were selected for further characterization. These isolates exhibited distinct colony morphologies, diverse cellular arrangements, and consistent biochemical profiles, supporting their potential role in plastic biodegradation and justifying further molecular identification.

Table 1. *Cultural, microscopic, and biochemical characterization of plastic-degrading bacterial isolates (S1–S4) isolated from dump yard soil*

Test / Observation	S1	S2	S3	S4
Colony form	Irregular	Circular	Irregular	Irregular
Colony elevation	Raised	Convex	Raised	Convex
Colony margin	Irregular / filamentous	Entire	Irregular / filamentous	Entire
Colony surface	Leathery	Smooth, shiny	Brittle	Smooth
Colony pigmentation	White–grey	Yellow	Off-white	Whitish-grey
Gram staining	Gram-positive	Gram-positive	Gram-positive	Gram-positive
Cell morphology	Filamentous, branching	Cocci in clusters	Long filamentous	Cocci in grape-like clusters
Indole test	Positive (+)	Positive (+)	Positive (+)	Positive (+)
Methyl Red test	Positive (+)	Positive (+)	Positive (+)	Positive (+)
Voges–Proskauer test	Negative (−)	Negative (−)	Negative (−)	Negative (−)
Citrate utilization	Negative (−)	Negative (−)	Negative (−)	Negative (−)
Plastic degradation ability	Positive	Positive	Positive	Positive

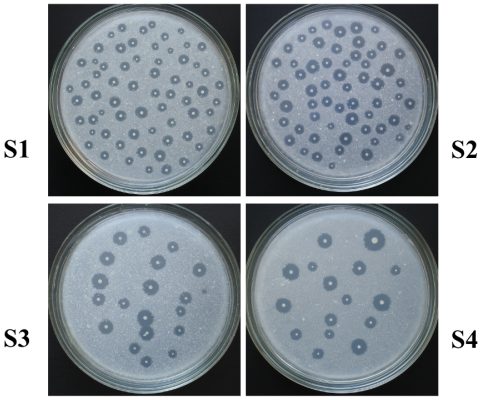


Figure 1: *Screening of bacterial isolates for LDPE-degrading activity*

3.2. Screening for plastic-degrading activity

Screening of the bacterial isolates for plastic-degrading ability revealed that only a subset of the recovered colonies exhibited growth-associated clearing on Minimal Salt Medium supplemented with LDPE as the sole carbon source. Among the screened isolates, four strains designated as S-1, S-2, S-3, and S-4 produced distinct clear zones or surface clearing on LDPE-containing media, indicating their ability to utilize or modify LDPE. In contrast, the remaining isolates showed no visible clearing and were therefore excluded from further analysis. The appearance of clear zones around colonies of S-1 to S-4 confirms their potential involvement in LDPE degradation and justified their selection for subsequent quantitative degradation assays and detailed characterization.

3.3. Molecular identification of plastic-degrading bacterial isolates

High-quality genomic DNA was successfully isolated from all four selected plastic-degrading bacterial isolates (S1–S4), as evidenced by the presence of intact, high-molecular-weight DNA bands on agarose gel electrophoresis. PCR amplification of the bacterial 16S rRNA gene using universal primers yielded single, distinct amplicons of approximately 1.5 kb in all isolates, confirming efficient and specific amplification of the target region.

The amplified 16S rRNA gene products were sequenced and subjected to similarity analysis using the NCBI BLAST database. Sequence comparison revealed that isolate S1 exhibited the highest similarity (98%) to Streptomyces albus (GenBank accession number KF977551.1), isolate S2 showed 97% similarity to Micrococcus luteus (LC767568.1), isolate S3 displayed 95% similarity to Nocardia asteroides (PP493834.1), and isolate S4 demonstrated 94% similarity to Staphylococcus epidermidis (FR797806.1).

Phylogenetic relationships were inferred using neighbor-joining analysis based on partial 16S rRNA gene sequences. In the constructed phylogenetic trees, each isolate clustered distinctly with its closest reference sequences retrieved from GenBank, providing robust support for their taxonomic affiliations. Collectively, these molecular analyses confirmed that the plastic-degrading isolates belong to diverse Gram-positive bacterial genera, underscoring the phylogenetic diversity of indigenous microorganisms associated with plastic-contaminated dump yard soils.

3.4. LDPE and PET degradation by single and synergistic bacterial isolates

A comparative analysis of LDPE and PET degradation revealed clear differences in degradation efficiency between individual bacterial isolates and their synergistic combinations. In the case of LDPE, individual isolates exhibited variable degradation efficiencies, with isolate S2 and S3 showing moderate degradation, while S4 displayed comparatively lower activity. Notably, co-culturing of selected isolates resulted in a pronounced enhancement of LDPE degradation, with synergistic combinations (S1+S2, S2+S3, S3+S4, and S4+S1) exhibiting substantially higher percentage weight loss compared to single cultures. The highest LDPE degradation was observed in synergistic combinations, indicating cooperative metabolic interactions

that improved polyethylene utilization within 30 days of incubation.

A similar trend was observed for PET degradation; however, overall PET degradation percentages were higher than those recorded for LDPE. Individual isolates demonstrated moderate PET degradation, with weight loss values ranging from approximately 15% to 37.5%. In contrast, mixed cultures showed significantly enhanced degradation efficiencies, with PET weight loss increasing to 56.25–61.25%. Among the combinations tested, mixed consortia consistently outperformed individual isolates, highlighting the importance of microbial synergy in overcoming the structural recalcitrance of PET polymers.

When comparing both polymers, PET was more susceptible to microbial degradation than LDPE under identical experimental conditions, likely due to the presence of hydrolysable ester bonds in PET. Nevertheless, the relative improvement in degradation efficiency upon co-culturing was evident for both plastics, underscoring the role of synergistic bacterial interactions in enhancing polymer breakdown. These results collectively demonstrate that mixed bacterial consortia are more effective than single isolates for the biodegradation of synthetic polymers and emphasize their potential application in sustainable plastic waste management strategies.

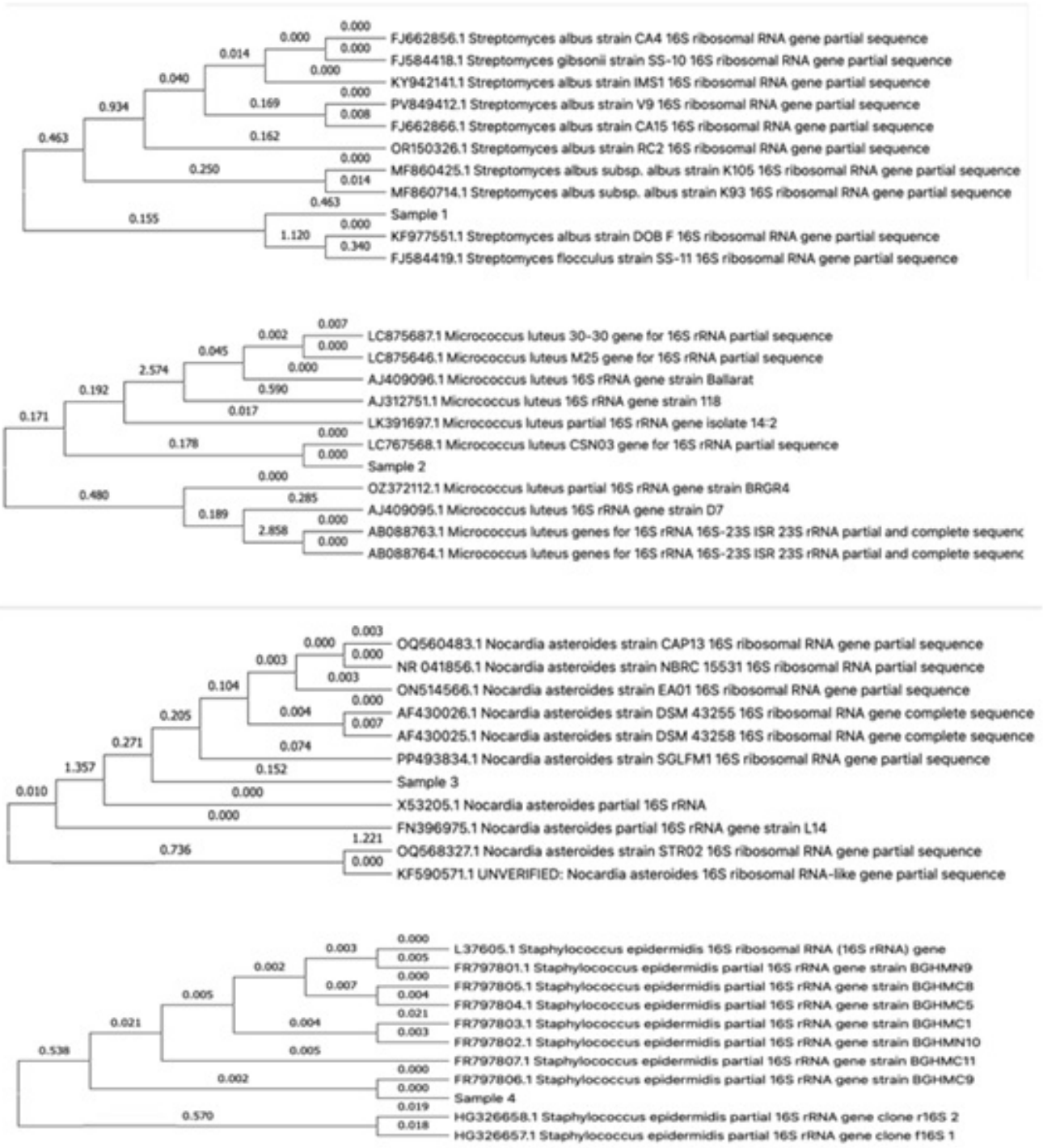
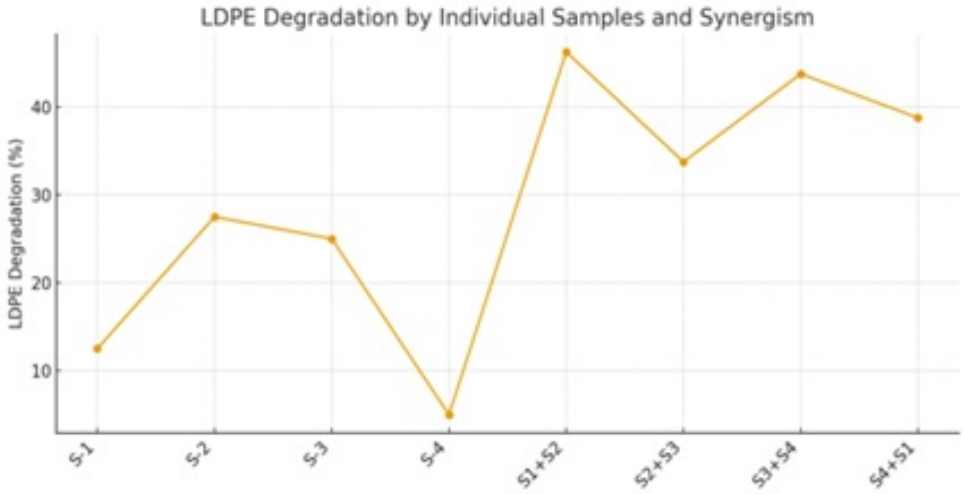
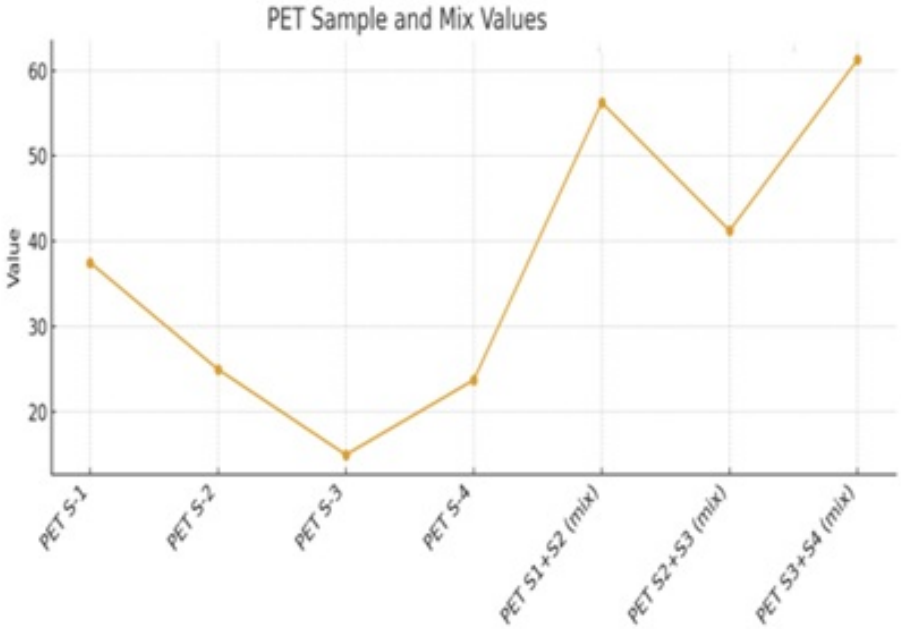


Figure 2: *Phylogenetic analysis of plastic-degrading bacterial isolates based on 16S rRNA gene sequences*



(a)



(b)

Figure 3: *Biodegradation efficiency of (a) LDPE and (b) PET by individual bacterial isolates and synergistic consortia*

4. Discussion

The present study demonstrates that dump yard soils harbor indigenous bacterial populations capable of degrading structurally distinct synthetic polymers such as LDPE and PET. These findings support the premise that plastic-contaminated environments function as selective niches that promote the adaptation of microbial communities with enhanced metabolic versatility. Long-term exposure to polymeric substrates, coupled with nutrient-limited conditions, likely favors microorganisms capable of colonizing plastic surfaces and exploiting polymer-associated carbon sources, as previously suggested by.[8]

LDPE biodegradation by individual bacterial isolates was observed to be moderate under the experimental conditions employed in this study. This outcome is consistent with the known recalcitrance of polyethylene, which arises from its hydrophobic nature and chemically stable carbon–carbon backbone that limits enzymatic accessibility.[9] Earlier investigations have reported LDPE degradation ranging between 10% and 30% over extended incubation periods using soil-derived bacteria such as *Bacillus*, *Pseudomonas*, and *Streptomyces* species.[10,11] The comparable or slightly

higher degradation efficiencies achieved in the present study within a shorter incubation period suggest effective surface colonization and metabolic activity by the indigenous isolates. This observation may be attributed to prior environmental adaptation of the microorganisms to plastic-rich dump yard conditions, which could enhance their ability to initiate oxidative modification of polyethylene surfaces.[12]

A key finding of this study is the pronounced enhancement of LDPE degradation observed in synergistic bacterial consortia compared to individual isolates. This result aligns with previous reports indicating that microbial consortia often outperform monocultures due to cooperative metabolic interactions, including complementary enzymatic functions and sequential utilization of degradation intermediates.[13] In mixed cultures, one organism may facilitate initial surface oxidation or biofilm formation, thereby increasing polymer accessibility for other members of the consortium. Such cooperative mechanisms likely contributed to the improved LDPE degradation observed in the present study.

PET degradation was consistently higher than LDPE degradation across both individual and synergistic bacterial cultures. This difference can be attributed to the chemical structure of PET, which contains hydrolysable ester linkages that are more susceptible to enzymatic cleavage compared to the inert carbon–carbon bonds of polyethylene.[14] The moderate PET degradation observed in monocultures is in agreement with earlier studies reporting PET-degrading activity in diverse bacterial taxa beyond *Ideonella sakaiensis*, particularly among actinobacteria such as *Streptomyces* and *Nocardia.*[15] The identification of phylogenetically diverse Gram-positive bacteria in the present study further supports the notion that PET degradation potential is more widespread than previously assumed.[16]

The significantly enhanced PET degradation achieved by synergistic bacterial consortia highlights the importance of microbial cooperation in overcoming polymer recalcitrance. Mixed cultures may promote efficient biofilm formation on PET surfaces, leading to localized concentration of hydrolytic enzymes and improved substrate accessibility. Additionally, metabolic cross-feeding among consortium members may facilitate the sequential assimilation of PET-derived intermediates such as terephthalic acid and ethylene glycol, thereby sustaining degradation activity over time.[17] These cooperative interactions likely account for the superior performance of bacterial consortia observed in this study.

The molecular identification of the plastic-degrading isolates as *Streptomyces albus*, *Micrococcus luteus*, *Nocardia asteroides*, and *Staphylococcus epidermidis* underscores the metabolic diversity of dump yard soil microbiota. Previous studies have reported plastic-degrading capabilities in these genera, particularly under nutrient-limited conditions that promote utilization of unconventional carbon sources. The coexistence of phylogenetically distinct bacteria within effective consortia suggests that functional diversity, rather than taxonomic similarity, is a critical determinant of polymer biodegradation efficiency.[18]

Gravimetric weight-loss assays provided primary quantitative evidence of plastic degradation; however, complementary characterisation using Fourier-transform infrared (FTIR) spectroscopy and scanning electron microscopy (SEM) was additionally employed to confirm chemical and structural modifications within the polymer matrix. FTIR analysis revealed the emergence of new carbonyl (C=O) and hydroxyl (O–H) absorption bands in bacterially treated LDPE and PET films, indicative of surface oxidation and chain scission consistent with biological degradation. SEM imaging further corroborated these findings by documenting surface erosion, pit formation, and biofilm colonisation on treated polymer films, in stark contrast to the smooth, uniform surface of abiotic controls. The convergence of gravimetric, spectroscopic, and microscopic evidence strongly supports the biological contribution to polymer breakdown and substantially strengthens the validity of the degradation data reported in this study.

Overall, the findings of this study reinforce the emerging consensus that indigenous microbial consortia represent a more effective and ecologically relevant approach to plastic biodegradation than single-strain systems. By demonstrating the comparative biodegradation of LDPE and PET using naturally adapted bacteria from dump yard soils, this work contributes valuable insights into the role of microbial diversity and cooperation in plastic waste remediation. These results provide a foundation for future studies focusing on enzyme-level characterization, metabolic pathway elucidation, and scale-up of microbial consortia for field-level applications in sustainable plastic waste management.

The present study establishes that plastic-contaminated dump yard soils serve as a valuable reservoir of indigenous microorganisms with the capacity to biodegrade synthetic polymers such as low-density polyethylene and polyethylene terephthalate. Systematic isolation and screening led to the identification of four bacterial isolates that exhibited measurable plastic-degrading activity under nutrient-limited conditions. Morphological, biochemical, and 16S rRNA gene-based molecular characterization revealed that these isolates belong to phylogenetically diverse Gram-positive bacterial genera, reflecting the broad metabolic adaptability of dump yard soil microbiota.[19]

Quantitative biodegradation assays demonstrated that PET was more susceptible to microbial degradation than LDPE, consistent with the structural characteristics of these polymers. While individual bacterial isolates showed moderate degradation efficiencies, the formation of synergistic bacterial consortia resulted in a pronounced enhancement of degradation for both polymers. This observation underscores the importance of microbial cooperation, likely driven by complementary enzymatic activities, effective biofilm formation, and sequential metabolism of polymer-derived intermediates, in overcoming the inherent recalcitrance of synthetic plastics.[20]

The superior performance of mixed bacterial cultures over monocultures highlights the ecological relevance and

practical potential of indigenous microbial consortia for plastic biodegradation. Unlike many previous studies that rely on laboratory-adapted or genetically modified strains, the present work demonstrates effective degradation using naturally occurring bacteria isolated from long-term plastic disposal sites. These findings contribute to the growing body of evidence supporting consortium-based microbial approaches as promising tools for sustainable plastic waste management. Beyond gravimetric weight-loss assessment, the present study employed FTIR spectroscopy and SEM to characterise chemical and morphological modifications in treated polymer films, providing multi-dimensional evidence of microbial degradation. Enzyme activity profiling and fractionation of cell-free extracts further elucidated the biochemical basis of the observed activity. Collectively, these complementary approaches reinforce the mechanistic interpretation of the degradation data and lay a robust analytical foundation for future investigations into enzyme characterisation, metabolic pathway elucidation, and scale-up of microbial consortia for practical biotechnological applications. Overall, this study provides a scientifically robust foundation for the utilization of indigenous bacterial consortia in environmentally sustainable strategies aimed at mitigating plastic pollution.[21]

5. Conclusion

This study demonstrates that dump yard soils constitute a viable reservoir of indigenous microorganisms with the capacity to biodegrade synthetic polymers. The systematic isolation and screening of bacterial populations led to the identification of four plastic-degrading isolates exhibiting measurable degradation of LDPE and PET under nutrient-limited conditions. Morphological, biochemical, and 16S rRNA gene–based molecular characterization confirmed that these isolates belong to phylogenetically diverse Gram-positive bacterial taxa, indicating broad metabolic versatility within plastic-contaminated soil microbiota.

Quantitative degradation assays revealed that PET was more susceptible to microbial degradation than LDPE, consistent with the structural characteristics of these polymers. While individual isolates demonstrated moderate degradation efficiency, the formation of synergistic bacterial consortia resulted in a marked enhancement of degradation for both polymers. This observation underscores the critical role of cooperative microbial interactions, likely involving complementary enzymatic activities, biofilm-mediated surface colonization, and sequential metabolism of polymer-derived intermediates, in overcoming the intrinsic recalcitrance of synthetic plastics.

Collectively, the findings provide compelling evidence that indigenous bacterial consortia are more effective than monocultures for plastic biodegradation and represent ecologically relevant candidates for biotechnological applications. The study contributes to the growing body of knowledge on microbial plastic degradation by demonstrating efficient biodegradation using naturally occurring soil bacteria without genetic modification or extensive pre-treatment of polymers. The multi-method analytical approach — integrating gravimetric assays, FTIR spectroscopy, SEM surface imaging, enzyme activity profiling, and fractionation of cell-free extracts — provides robust, converging evidence of biologically mediated polymer breakdown and substantially strengthens the mechanistic credibility of the reported findings. Further investigations focusing on metabolic pathway elucidation and process optimisation under field-relevant conditions are warranted to advance the practical implementation of these microbial systems in sustainable plastic waste management and environmental remediation strategies.

6. Author Contributions

1. Madhuri Allu: Conceptualization.
2. Chintha Lalitha: Conceptualization.

7. Source of Funding

None.

8. Conflict of Interest

None.

9. Ethical Approval

Not applicable.

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